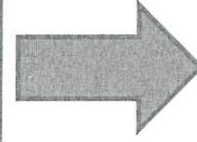
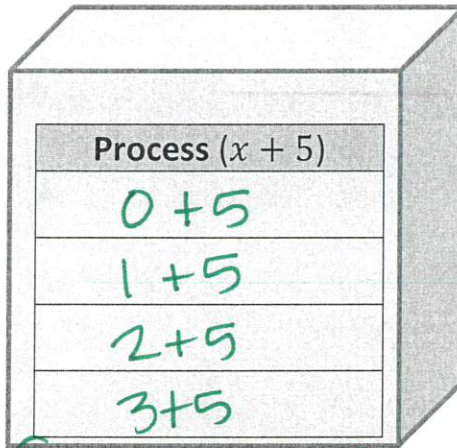
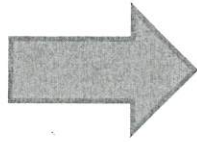


Input/Output Machine

Independent

Input (x)
0
1
2
3



Dependent

Output (y)
5
6
7
8

Practice:

Input x	Process	Output y
0	$4(0)$	0
2	$4(2)$	8
4	$4(4)$	16
6	$4(6)$	24

Equation: $y = 4x$

Multiplicative

$$y = ax \quad a = 4$$

or

Additive

$$y = x + a \quad a =$$

Input x	Process	Output y
0	$0 - 2$	-2
1	$1 - 2$	-1
2	$2 - 2$	0
3	$3 - 2$	1

Equation: $y = x - 2$

Multiplicative

$$y = ax \quad a =$$

or

Additive

$$y = x + a \quad a = -2$$

Input, x	0	1	3	5
Process	$0 + 7$	$1 + 7$	$3 + 7$	$5 + 7$
Output, y	7	8	10	12

Equation: $y = x + 7$

Multiplicative

$$y = ax \quad a =$$

or

Additive

$$y = x + a \quad a = 7$$

Input, x	0	1	2	3
Process	$3(0)$	$3(1)$	$3(2)$	$3(3)$
Output, y	0	3	6	9

Equation: $y = 3x$

Multiplicative

$$y = ax \quad a = 3$$

or

Additive

$$y = x + a \quad a =$$

1. The table below shows the number of miles Erica can drive with different amounts of gasoline. Which equation correctly shows the relationship between the amount of gasoline Erica has and the number of miles she can drive?

Gallons of Gas (g)	Number of Miles (m)
6 $\times 25$	150
8 $\times 25$	200
10 $\times 25$	250
12 $\times 25$	300

A. $m = g + 50$

B. $m = \frac{1}{25}g$

C. $m = g + 2$

D. $m = 25g$

$m = 25g$

2. David saves his weekly allowance. The table below shows how much total money he has saved after different numbers of weeks. Which equation correctly shows the relationship between the number of weeks and the amount of money David saved?

Weeks (w)	Money Saved (m)
1 $\times 5$	5
2 $\times 5$	10
3 $\times 5$	15
4 $\times 5$	20

A. $m = 2w$

B. $m = 5w$

C. $m = w + 1$

D. $m = w + 5$

$m = 5(w)$

3. The numbers of boys and girls in Ms. Lambert's classes are shown in the table below. Which equation correctly shows the relationship between the number of girls and the number of boys in each of Ms. Smith's classes?

	Girls (g)	Boys (b)
Class 1	6 $+ 5$	11
Class 2	8 $+ 5$	13
Class 3	10 $+ 5$	15
Class 4	12 $+ 5$	17

A. $b = g + 5$

B. $b = g - 5$

C. $b = g + 2$

D. $b = g - 2$

$b = g + 5$